

FURTHER ANALOGUES OF ANDREWS' SPT-FUNCTION IDENTITY

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Abstract

In this note, we give new analogues of Andrews' celebrated spt-function identity using basic hypergeometric transformations due to Verma and Jain.

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1. Introduction

In 2008, Andrews introduced the world to a weighted count on partitions given by counting a partition by the number of times the smallest part appears. This function, denoted by $\text{spt}(n)$, and defined as the number of smallest parts among all the partitions of n is known to possess an intriguing modular structure and satisfy the Ramanujan-type congruences

$$\text{spt}(5n + 4) \equiv 0 \pmod{5},$$

$$\text{spt}(7n + 5) \equiv 0 \pmod{7},$$

$$\text{spt}(13n + 6) \equiv 0 \pmod{13},$$

among other beautiful combinatorial and arithmetic properties. The origin of these arithmetic properties is attributed to the connection between its generating function and harmonic Maass forms.

Let N, m, n be non-negative integers. Throughout we use the following standard notations.

$$(a)_N = (a; q)_N := \prod_{k=0}^{N-1} (1 - aq^k),$$

$$(a)_\infty = (a; q)_\infty := \lim_{N \rightarrow \infty} (a)_N \text{ where } |q| < 1.$$

The following definition of the basic hypergeometric series ${}_{r+1}\phi_r$ will be used throughout

$${}_{r+1}\phi_r \left(\begin{matrix} a_1, a_2, \dots, a_{r+1} \\ b_1, b_2, \dots, b_r \end{matrix}; q, z \right) := \sum_{n=0}^{\infty} \frac{(a_1; q)_n (a_2; q)_n \cdots (a_{r+1}; q)_n}{(q; q)_n (b_1; q)_n (b_2; q)_n \cdots (b_r; q)_n} z^n.$$

When we do not specify the base as in $(a)_n$, the base is q .

By standard combinatorial arguments, the generating function of the smallest parts function (spt-function) due to Andrews is given by

$$\sum_{n=1}^{\infty} \text{spt}(n) q^n = \sum_{n=1}^{\infty} \frac{q^n}{(1 - q^n)^2 (q^{n+1}; q)_{\infty}}.$$

Starting from this, Andrews' identity for the spt-function, which relates the spt-function to the partition function and the second Atkin-Garvan rank moment, is obtained using Watson's q -analogue of Whipple's theorem (${}_8\phi_7$ -transformation)

$$\begin{aligned} & {}_8\phi_7 \left[\begin{matrix} a, q\sqrt{a}, -q\sqrt{a}, b, c, d, e, q^{-N} \\ \sqrt{a}, -\sqrt{a}, aq/b, aq/c, aq/d, aq/e, aq^{N+1}; q, \frac{a^2 q^{N+1}}{bcde} \end{matrix} \right] \\ &= \frac{(aq, aq/de; q)_N}{(aq/d, aq/e; q)_N} {}_4\phi_3 \left[\begin{matrix} aq/bc, d, e, q^{-N} \\ aq/b, aq/c, deq^{-N}/a; q, q \end{matrix} \right]. \end{aligned}$$

This spt-function identity of Andrews [3, Theorem 4] is given by

$$\sum_{n=1}^{\infty} \text{spt}(n) q^n = \frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{nq^n}{1 - q^n} + \frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{(-1)^n (1 + q^n)}{(1 - q^n)^2} q^{\frac{3n}{2}(n+1)}. \quad (1.1)$$

The first sum on the right hand side is the generating function for $np(n)$, where $p(n)$ denotes the number of partitions of n and the second is the generating function for $\frac{1}{2}N_2(n)$, where

$$N_2(n) = \sum_{m=-\infty}^{\infty} m^2 N(m, n)$$

is the second Atkin-Garvan rank moment. Here $N(m, n)$ denotes the number of partitions of n with rank m . The rank of a partition is defined to be the largest part minus the number of parts. For instance, the rank of the partition $5 + 3 + 3 + 1$ is $5 - 4 = 1$. See [8, 10] for more on the rank and [7] for more on rank moments.

In 2015, Andrews, Dixit and Yee wrote a paper [4] where they obtained new partition theoretic interpretations for $q\omega(q)$ and $\nu(-q)$ where $\omega(q)$ and $\nu(q)$ are the third order mock theta functions

$$\omega(q) = \sum_{n=0}^{\infty} \frac{q^{2n(n+1)}}{(q; q^2)_{n+1}},$$

and

$$\nu(q) = \sum_{n=0}^{\infty} \frac{q^{n(n+1)}}{(-q; q^2)_{n+1}},$$

among a wide range of other results, perhaps the most striking of which are the congruences

$$\text{spt}_{\omega}(5n+3) \equiv 0 \pmod{5},$$

$$\text{spt}_{\omega}(10n+7) \equiv 0 \pmod{5},$$

$$\text{spt}_{\omega}(10n+9) \equiv 0 \pmod{5},$$

where $\text{spt}_{\omega}(n)$ is the total number of smallest parts in all of the partitions of n in which each odd part is less than twice the smallest part. These three divisibility properties are a consequence of a q -series identity obtained using the following limiting case of a $_{10}\phi_9$ -transformation due to Bailey (limiting case due to Andrews [2, Equation 2.10])

$$\begin{aligned} \lim_{N \rightarrow \infty} {}_{10}\phi_9 \left(a, q^2 \sqrt{a}, -q^2 \sqrt{a}, p_1, p_1 q, p_2, p_2 q, f, q^{-2N}, q^{-2N+1}, \frac{a^3 q^{4N+3}}{p_1^2 p_2^2 f} \right) \\ = \frac{(aq; q)_{\infty} \left(\frac{aq}{p_1 p_2}; q \right)_{\infty}}{\left(\frac{aq}{p_1}; q \right)_{\infty} \left(\frac{aq}{p_2}; q \right)_{\infty}} \sum_{n=0}^{\infty} \frac{(p_1; q)_n (p_2; q)_n \left(\frac{aq}{f}; q^2 \right)_n}{(q; q)_n (aq; q^2)_n \left(\frac{aq}{f}; q \right)_n} \left(\frac{aq}{p_1 p_2} \right)^n. \end{aligned} \quad (1.2)$$

The aforementioned q -series identity due to Andrews, Dixit and Yee [4, Lemma 6.1] is an analogue of the Andrews spt-function identity (1.1) and is given by

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{q^n}{(1-q^n)^2 (q^{n+1}; q)_{\infty} (q^{2n+2}; q^2)_{\infty}} \\ = \frac{1}{(q^2; q^2)_{\infty}} \sum_{n=1}^{\infty} \frac{nq^n}{1-q^n} + \frac{1}{(q^2; q^2)_{\infty}} \sum_{n=1}^{\infty} \frac{(-1)^n (1+q^{2n})}{(1-q^{2n})^2} q^{n(3n+1)}. \end{aligned} \quad (1.3)$$

Now, in a recent paper, Andrews and Bachraoui established new partition theoretic interpretations which state that $q\omega(q)$ and $\nu(-q)$ are generating functions for two-color partitions in which the even or the odd parts may occur only in one color, without appeal to such analogues. And, even more recently, a newly discovered analogue of Andrews spt-function identity by Andrews and Kumar [6, Theorem 1.3] has led them to discover a new partition theoretic interpretation of the other third order mock theta function

$$f(q) = \sum_{n=0}^{\infty} \frac{q^{n^2}}{(-q; q)_n^2},$$

which in turn relates partitions with positive odd rank and two-color partitions [6, Theorems 1.1 and 1.2]. This new analogue is obtained using the limiting case of another $_{10}\phi_9$ -transformation due to Bailey (limiting case due to Andrews [2, Equation 2.9])

$$\begin{aligned} & \lim_{N \rightarrow \infty} {}_{10}\phi_9 \left(\begin{matrix} a, q\sqrt{a}, -q\sqrt{a}, b, r_1, -r_1, r_2, -r_2, q^{-N}, -q^{-N} \\ \sqrt{a}, -\sqrt{a}, \frac{aq}{b}, \frac{aq}{r_1}, -\frac{aq}{r_1}, \frac{aq}{r_2}, -\frac{aq}{r_2}, aq^{N+1}, -aq^{N+1}; q, -\frac{a^3q^{3+2N}}{br_1^2r_2^2} \end{matrix} \right) \\ &= \frac{(a^2q^2; q^2)_\infty \left(\frac{a^2q^2}{r_1^2r_2^2}; q^2 \right)_\infty}{\left(\frac{a^2q^2}{r_1^2}; q^2 \right)_\infty \left(\frac{a^2q^2}{r_2^2}; q^2 \right)_\infty} \sum_{n=0}^{\infty} \frac{(r_1^2; q^2)_n (r_2^2; q^2)_n (-aq/b; q)_{2n}}{(q^2; q^2)_n (a^2q^2/b^2; q^2)_n (-aq; q)_{2n}} \left(\frac{a^2q^2}{r_1^2r_2^2} \right)^n, \end{aligned} \quad (1.4)$$

and is given by

$$\begin{aligned} & \sum_{n=1}^{\infty} \frac{q^{2n}}{(1 - q^{2n})^2 (-q^{n+1}; q)_\infty (q^{n+1}; q)_\infty} \\ &= \frac{1}{(q; q)_\infty} \sum_{n=1}^{\infty} \frac{nq^{2n}}{1 - q^{2n}} + \frac{1}{(q; q)_\infty} \sum_{n=1}^{\infty} \frac{(-1)^n (1 + q^n)}{(1 - q^{2n})^2} q^{\frac{3n}{2}(n+1)}. \end{aligned} \quad (1.5)$$

As noted above, $_{10}\phi_9$ -transformations have been used to produce such analogues so far. In this paper, we derive more such analogues using higher parameter basic hypergeometric transformations.

We will first consider the following $_{12}\phi_{11}$ -transformation due to Verma and Jain [13, Equation 1.5], [12, Equation 3.5] to arrive at another analogue of Andrews' spt-function identity.

$$\begin{aligned} & {}_{12}\phi_{11} \left(\begin{matrix} a, q\sqrt{a}, -q\sqrt{a}, x, \omega x, \omega^2 x, y, \omega y, \omega^2 y, q^{-N}, \omega q^{-N}, \omega^2 q^{-N} \\ \sqrt{a}, -\sqrt{a}, \frac{aq}{x}, \frac{aq}{\omega x}, \frac{aq}{\omega^2 x}, \frac{aq}{y}, \frac{aq}{\omega y}, \frac{aq}{\omega^2 y}, aq^{N+1}, \omega^2 aq^{N+1}, \omega aq^{N+1}; q, \frac{a^4 q^{4+3N}}{x^3 y^3} \end{matrix} \right) \\ &= \frac{(a^3 q^3; q^3)_N (a^3 q^3/x^3 y^3; q^3)_N}{(a^3 q^3/x^3; q^3)_N (a^3 q^3/y^3; q^3)_N} {}_6\phi_5 \left(\begin{matrix} q^{-3N}, x^3, y^3, aq, aq^2, aq^3 \\ (aq)^{\frac{3}{2}}, -(aq)^{\frac{3}{2}}, a^{\frac{3}{2}} q^3, -a^{\frac{3}{2}} q^3, \frac{x^3 y^3 q^{-3N}}{a^3}; q^3, q^3 \end{matrix} \right). \end{aligned} \quad (1.6)$$

This identity is one of several such identities between basic hypergeometric functions on different bases developed by the authors in [13] leading to a number of new identities of the Rogers-Ramanujan type.

Remark 1. There is a minor error in this identity in Sills' paper [12, Equation 3.5] where a minus sign seems to appear before the last parameter $\frac{a^4 q^{4+3N}}{x^3 y^3}$ on the left hand side.

Theorem 1.1 (First analogue of Andrews' spt-function identity).

$$\sum_{n=1}^{\infty} \frac{q^{3n}}{(1-q^{3n})^2(q^{3(n+1)}; q^3)_n(q^{3n+1}; q)_{\infty}} = \frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{nq^{3n}}{1-q^{3n}} + \frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{(-1)^n(1+q^n)}{(1-q^{3n})^2} q^{\frac{n(3n+5)}{2}}.$$

Next we will consider the other ${}_{12}\phi_{11}$ -transformation due to Verma and Jain [13, Equation 1.6], [12, Equation 3.6] stated below to arrive at yet another analogue of Andrews' spt-function identity.

$$\begin{aligned} & {}_{12}\phi_{11} \left(\begin{matrix} a, q^3 \sqrt{a}, -q^3 \sqrt{a}, x, xq, xq^2, y, yq, yq^2, q^{-N}, q^{-N+1}, q^{-N+2} \\ \sqrt{a}, -\sqrt{a}, \frac{aq^3}{x}, \frac{aq^2}{x}, \frac{aq}{x}, \frac{aq^3}{y}, \frac{aq^2}{y}, \frac{aq}{y}, aq^{N+3}, aq^{N+2}, aq^{N+1} \end{matrix}; q^3; \frac{a^4 q^{3+3N}}{x^3 y^3} \right) \\ &= \frac{(aq; q)_N (aq/xy; q)_N}{(aq/x; q)_N (aq/y; q)_N} {}_6\phi_5 \left(\begin{matrix} \sqrt[3]{a}, \omega \sqrt[3]{a}, \omega^2 \sqrt[3]{a}, x, y, q^{-n} \\ \sqrt{a}, -\sqrt{a}, \sqrt{aq}, -\sqrt{aq}, \frac{xy}{a} q^{-n} \end{matrix}; q; q \right). \end{aligned} \quad (1.7)$$

Theorem 1.2 (Second analogue of Andrews' spt-function identity).

$$\sum_{n=1}^{\infty} \frac{q^n}{(1-q^n)(q^{3n}; q^3)_{\infty}(-q; q)_{n-1}(q; q^2)_n} = \frac{1}{(q^3; q^3)_{\infty}} \sum_{n=1}^{\infty} \frac{nq^n}{1-q^n} + \frac{1}{(q^3; q^3)_{\infty}} \sum_{n=1}^{\infty} \frac{(-1)^n(1+q^{3n})}{(1-q^{3n})^2} q^{\frac{3n(3n+1)}{2}}.$$

The paper contains two more sections. Section 2 is devoted to proofs of the two theorems stated above. In the final section, we discuss possible questions arising from our work.

2. Proof of the theorems

2.1. Proof of Theorem 1.1

PROOF. By [6, page 7], for $n \geq 1$, we know

$$\lim_{a \rightarrow 1} \frac{(a; q)_n (q \sqrt{a}; q)_n (-q \sqrt{a}; q)_n}{(\sqrt{a}; q)_n (-\sqrt{a}; q)_n} = (q; q)_n (1 + q^n).$$

Then, letting $a \rightarrow 1$, $x = z$, $y = z^{-1}$ in Equation (1.6) we have

$$\begin{aligned} & 1 + \sum_{n=1}^{\infty} \frac{(1+q^n)(z)_n(\omega z)_n(\omega^2 z)_n(z^{-1})_n(\omega z^{-1})_n(\omega^2 z^{-1})_n(q^{-N})_n(\omega q^{-N})_n(\omega^2 q^{-N})_n}{(q/z)_n(q/\omega z)_n(q/\omega^2 z)_n(qz)_n(qz/\omega)_n(qz/\omega^2)_n(q^{N+1})_n(\omega^2 q^{N+1})_n(\omega q^{N+1})_n} q^{3Nn+4n} \\ &= \frac{(q^3; q^3)_N^2}{(\frac{q^3}{z^3}; q^3)_N (q^3 z^3; q^3)_N} \sum_{n=0}^{\infty} \frac{(q^{-3N}; q^3)_n (z^3; q^3)_n (z^{-3}; q^3)_n (q; q^3)_n (q^2; q^3)_n (q^3; q^3)_n}{(q^3; q^3)_n (q^{\frac{3}{2}}; q^3)_n (-q^{\frac{3}{2}}; q^3)_n (q^3; q^3)_n (-q^3; q^3)_n (q^{-3N}; q^3)_n} q^{3n}. \end{aligned}$$

Now,

$$\lim_{N \rightarrow \infty} \frac{(q^{-N})_n (\omega q^{-N})_n (\omega^2 q^{-N})_n}{(q^{N+1})_n (\omega^{-1} q^{N+1})_n (\omega^{-2} q^{N+1})_n} q^{3Nn} = (-1)^n q^{\frac{3n(n-1)}{2}}.$$

Then, letting $N \rightarrow \infty$ in the identity above and simplifying, we get

$$\begin{aligned} 1 + \sum_{n=1}^{\infty} \frac{(-1)^n (1+q^n) (z^3; q^3)_n (z^{-3}; q^3)_n}{(z^{-3} q^3; q^3)_n (z^3 q^3; q^3)_n} q^{\frac{n(3n+5)}{2}} \\ = \frac{(q^3; q^3)_{\infty}^2}{(z^{-3} q^3; q^3)_{\infty} (z^3 q^3; q^3)_{\infty}} \sum_{n=0}^{\infty} \frac{(z^3; q^3)_n (z^{-3}; q^3)_n (q; q)_{3n}}{(q^3; q^6)_n (q^6; q^6)_n (q^3; q^3)_n} q^{3n}. \end{aligned}$$

Thus

$$\begin{aligned} 1 + \sum_{n=1}^{\infty} \frac{(-1)^n (1+q^n) (1-z^3) (1-z^{-3})}{(1-z^3 q^{3n}) (1-z^{-3} q^{3n})} q^{\frac{n(3n+5)}{2}} \\ = \frac{(q^3; q^3)_{\infty}^2}{(z^{-3} q^3; q^3)_{\infty} (z^3 q^3; q^3)_{\infty}} \sum_{n=0}^{\infty} \frac{(z^3; q^3)_n (z^{-3}; q^3)_n (q; q)_{3n}}{(q^3; q^3)_{2n} (q^3; q^3)_n} q^{3n}. \end{aligned}$$

Replacing z^3 by z and rearranging, we obtain

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{(z; q^3)_n (z^{-1}; q^3)_n (q; q)_{3n}}{(q^3; q^3)_{2n} (q^3; q^3)_n} q^{3n} \\ = \frac{1}{(q^3; q^3)_{\infty}^2} \left((zq^3; q^3)_{\infty} (z^{-1} q^3; q^3)_{\infty} + \sum_{n=1}^{\infty} \frac{(-1)^n (1+q^n) (z; q^3)_{\infty} (z^{-1}; q^3)_{\infty}}{(1-zq^{3n}) (1-z^{-1} q^{3n})} q^{\frac{n(3n+5)}{2}} \right). \end{aligned} \quad (2.1)$$

Next, we use the result [3, Equation (2.1)]

$$-\frac{1}{2} \frac{d^2}{dz^2} \left((1-z)(1-z^{-1})f(z) \right) = f(1)$$

to differentiate the following functions twice with respect to z . First we consider

$$f(z) = (zq^3, q^3)_{n-1} (z^{-1} q^3, q^3)_{n-1}$$

resulting in

$$-\frac{1}{2} \frac{d^2}{dz^2} \left((z; q^3)_n (z^{-1}; q^3)_n \right) \Big|_{z=1} = (q^3; q^3)_{n-1}^2$$

and then consider

$$f(z) = \frac{(zq^3, q^3)_{\infty} (z^{-1} q^3, q^3)_{\infty}}{(1-zq^{3n})(1-z^{-1} q^{3n})},$$

resulting in

$$-\frac{1}{2} \frac{d^2}{dz^2} \left(\frac{(z; q^3)_\infty (z^{-1}; q^3)_\infty}{(1 - zq^{3n})(1 - z^{-1}q^{3n})} \right) \Big|_{z=1} = \frac{(q^3; q^3)_\infty^2}{(1 - q^{3n})^2}.$$

Finally, replacing $q \rightarrow q^3$ in [3, Equation (2.4)] we obtain

$$-\frac{1}{2} \frac{d^2}{dz^2} \left((zq^3; q^3)_\infty (z^{-1}q^3; q^3)_\infty \right) \Big|_{z=1} = (q^3; q^3)_\infty^2 \sum_{n=1}^{\infty} \frac{nq^{3n}}{1 - q^{3n}}.$$

Differentiating both sides of (2.1) twice with respect to z and using the above three results, we get

$$\sum_{n=1}^{\infty} \frac{(q^3; q^3)_{n-1}^2 (q; q)_{3n}}{(q^3; q^3)_{2n} (q^3; q^3)_n} q^{3n} = \sum_{n=1}^{\infty} \frac{nq^{3n}}{1 - q^{3n}} + \sum_{n=1}^{\infty} \frac{(-1)^n (1 + q^n)}{(1 - q^{3n})^2} q^{\frac{n(3n+5)}{2}},$$

from where the theorem is obtained by dividing throughout by $(q; q)_\infty$ and rewriting the left hand side. $\square$

2.2. Proof of Theorem 1.2

PROOF. As in the proof of Theorem 1.1, for $n \geq 1$, we have

$$\lim_{a \rightarrow 1} \frac{(a; q^3)_n (q^3 \sqrt{a}; q^3)_n (-q^3 \sqrt{a}; q^3)_n}{(\sqrt{a}; q^3)_n (-\sqrt{a}; q^3)_n} = (q^3; q^3)_n (1 + q^{3n}).$$

Also, for $n \geq 1$, we have

$$\lim_{a \rightarrow 1} \frac{(\sqrt[3]{a}; q)_n (\omega \sqrt[3]{a}; q)_n (\omega^2 \sqrt[3]{a}; q)_n}{(\sqrt{a}; q)_n (-\sqrt{a}; q)_n} = \frac{(q^3; q^3)_{n-1}}{(q^2; q^2)_{n-1}}.$$

Then, letting $a \rightarrow 1$, $x = z$, $y = z^{-1}$ in Equation (1.7) we have

$$\begin{aligned} & 1 + \sum_{n=1}^{\infty} \frac{(1 + q^{3n})(z; q^3)_n (zq; q^3)_n (zq^2; q^3)_n (z^{-1}; q^3)_n (z^{-1}q; q^3)_n (z^{-1}q^2; q^3)_n (q^{-N}; q^3)_n (q^{-N+1}; q^3)_n (q^{-N+2}; q^3)_n}{\left(\frac{q^3}{z}; q^3\right)_n \left(\frac{q^2}{z}; q^3\right)_n \left(\frac{q}{z}; q^3\right)_n (zq^3; q^3)_n (zq^2; q^3)_n (zq; q^3)_n (q^{N+3}; q^3)_n (q^{N+2}; q^3)_n (q^{N+1}; q^3)_n} \\ & \quad \cdot q^{3Nn+3n} \\ & = \frac{(q; q)_N^2}{(q/z; q)_N (qz; q)_N} + \frac{(q; q)_N^2}{(q/z; q)_N (qz; q)_N} \sum_{n=1}^{\infty} \frac{(q^3; q^3)_{n-1} (z; q)_n (z^{-1}; q)_n}{(q; q)_n (q^2; q^2)_{n-1} (\sqrt{q}; q)_n (-\sqrt{q}; q)_n} q^n. \end{aligned}$$

Now,

$$\lim_{N \rightarrow \infty} \frac{(q^{-N}; q^3)_n (q^{-N+1}; q^3)_n (q^{-N+2}; q^3)_n}{(q^{N+3}; q^3)_n (q^{N+2}; q^3)_n (q^{N+1}; q^3)_n} q^{3Nn} = (-1)^n q^{\frac{3n(3n-1)}{2}}.$$

Then, letting $N \rightarrow \infty$ in the identity above and simplifying, we get

$$\begin{aligned} 1 + \sum_{n=1}^{\infty} \frac{(-1)^n (1 + q^{3n})(z; q)_{3n}(z^{-1}; q)_{3n}}{(z^{-1}q; q)_{3n}(zq; q)_{3n}} q^{\frac{3n(3n+1)}{2}} \\ = \frac{(q; q)_{\infty}^2}{(z^{-1}q; q)_{\infty}(zq; q)_{\infty}} + \frac{(q; q)_{\infty}^2}{(z^{-1}q; q)_{\infty}(zq; q)_{\infty}} \sum_{n=1}^{\infty} \frac{(z; q)_n(z^{-1}; q)_n(q^3; q^3)_{n-1}}{(q; q)_n(q^2; q^2)_{n-1}(q; q^2)_n} q^n. \end{aligned}$$

Thus

$$\begin{aligned} 1 + \sum_{n=1}^{\infty} \frac{(-1)^n (1 + q^{3n})(1 - z)(1 - z^{-1})}{(1 - zq^{3n})(1 - z^{-1}q^{3n})} q^{\frac{3n(3n+1)}{2}} \\ = \frac{(q; q)_{\infty}^2}{(z^{-1}q; q)_{\infty}(zq; q)_{\infty}} + \frac{(q; q)_{\infty}^2}{(z^{-1}q; q)_{\infty}(zq; q)_{\infty}} \sum_{n=1}^{\infty} \frac{(z; q)_n(z^{-1}; q)_n(q^3; q^3)_{n-1}}{(q; q)_n(q^2; q^2)_{n-1}(q; q^2)_n} q^n. \end{aligned}$$

Rearranging, we obtain

$$\begin{aligned} 1 + \sum_{n=1}^{\infty} \frac{(z; q)_n(z^{-1}; q)_n(q^3; q^3)_{n-1}}{(q; q)_n(q^2; q^2)_{n-1}(q; q^2)_n} q^n \\ = \frac{1}{(q; q)_{\infty}^2} \left((zq; q)_{\infty}(z^{-1}q; q)_{\infty} + \sum_{n=1}^{\infty} \frac{(-1)^n (1 + q^{3n})(z; q)_{\infty}(z^{-1}; q)_{\infty}}{(1 - zq^{3n})(1 - z^{-1}q^{3n})} q^{\frac{3n(3n+1)}{2}} \right). \end{aligned} \quad (2.2)$$

Next, like in the proof of the previous theorem, we again use the result

$$-\frac{1}{2} \frac{d^2}{dz^2} \left((1 - z)(1 - z^{-1})f(z) \right) = f(1)$$

to differentiate the following functions twice with respect to z . First, considering

$$f(z) = (zq, q)_{n-1}(z^{-1}q, q)_{n-1}$$

results in the following double derivative also found in Andrews original work [3, Proof of Theorem 3]

$$-\frac{1}{2} \frac{d^2}{dz^2} \left((z; q)_n(z^{-1}; q)_n \right) \Big|_{z=1} = (q; q)_{n-1}^2,$$

and then considering

$$f(z) = \frac{(zq, q)_{\infty}(z^{-1}q, q)_{\infty}}{(1 - zq^{3n})(1 - z^{-1}q^{3n})}$$

results in

$$-\frac{1}{2} \frac{d^2}{dz^2} \left(\frac{(z; q^3)_{\infty}(z^{-1}; q^3)_{\infty}}{(1 - zq^{3n})(1 - z^{-1}q^{3n})} \right) \Big|_{z=1} = \frac{(q; q)_{\infty}^2}{(1 - q^{3n})^2}.$$

Finally, [3, Equation (2.4)] in Andrews original work states

$$-\frac{1}{2} \frac{d^2}{dz^2} \left((zq; q)_\infty (z^{-1}q; q)_\infty \right) \Big|_{z=1} = (q; q)_\infty^2 \sum_{n=1}^{\infty} \frac{nq^n}{1-q^n}.$$

Differentiating both sides of (2.2) twice with respect to z and using the above three results, we get

$$\sum_{n=1}^{\infty} \frac{(q; q)_{n-1}^2 (q^3; q^3)_{n-1}}{(q; q)_n (q^2; q^2)_{n-1} (q; q^2)_n} q^n = \sum_{n=1}^{\infty} \frac{nq^n}{1-q^n} + \sum_{n=1}^{\infty} \frac{(-1)^n (1+q^{3n})}{(1-q^{3n})^2} q^{\frac{3n(3n+1)}{2}},$$

from where the theorem is obtained by dividing throughout by $(q^3; q^3)_\infty$ and rewriting the left hand side. $\square$

3. Concluding observations

Several possible avenues for further investigation arise from our work.

1. In the works of Andrews, Dixit and Yee [4] and that of Andrews and Kumar [6], the spt-function identities have subsequently led to connections with third order mock theta functions discussed earlier in the introduction section. In both the cases, this connection is established using a non-weighted version of the sum on the left hand side of the identities in Equations (1.3) and (1.5) and working on this closely related series after initial use of the following four-parameter deep identity due to Andrews [1, Theorem 1]

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{(B; q)_n (-Abq; q)_n q^n}{(-aq; q)_n (-bq; q)_n} &= \frac{-a^{-1}(B; q)_\infty (-Abq; q)_\infty}{(-bq; q)_\infty (-aq; q)_\infty} \sum_{m=0}^{\infty} \frac{(A^{-1}; q)_m}{(-B/a; q)_{m+1}} \left(\frac{Abq}{a} \right)^m \\ &+ (1+b) \sum_{m=0}^{\infty} \frac{(-a^{-1}; q)_{m+1} (-ABq/a; q)_m (-b)^m}{(-B/a; q)_{m+1} (Abq/a; q)_{m+1}}. \end{aligned} \quad (3.1)$$

For instance, it is shown in [4, Theorem 3.1] that

$$q\omega(q) = \sum_{n=1}^{\infty} \frac{q^n}{(1-q^n)(q^{n+1}; q)_n (q^{2n+2}; q)_\infty}.$$

It is possible that such mock theta connections or relations with other partition theoretic objects also exist for our spt-function identities. One attempt at this by

removing the quadratic factor in the denominator of the identity in Theorem 1.1 leads to the following simplification

$$\begin{aligned}
 & \sum_{n=1}^{\infty} \frac{q^{3n}}{(1 - q^{3n})(q^{3(n+1)}; q^3)_n (q^{3n+1}; q)_{\infty}} \\
 &= \sum_{n=0}^{\infty} \frac{q^{3n+3}}{(q^{3n+3}; q^3)_{n+2} (q^{3n+4}; q)_{\infty}} \\
 &= \frac{q^3}{(q; q)_{\infty}} \sum_{n=0}^{\infty} \frac{(q; q)_{3n+3}}{(q^{3n+3}; q^3)_{n+2}} q^{3n} \\
 &= \frac{q^3}{(q; q)_{\infty}} \sum_{n=0}^{\infty} \frac{(q^3; q^3)_n (q; q^3)_{n+1} (q^2; q^3)_{n+1}}{(q^3; q^6)_{n+1} (-q^3; q^3)_{n+1}} q^{3n}.
 \end{aligned}$$

Use of Andrews' identity (3.1) does not seem useful here due to presence of more parameters in the series above. We believe this can be dealt with by considering a higher parameter identity analogous to Andrews. Some instances of such identities found in literature are [5, Theorem 1.1] and [9, Theorem 1.3]. Also, our investigation here is preliminary and a different approach to simplifying and transforming the series is worth considering.

2. There are more such analogues to be discovered. Some identities that could possibly yield such analogues are [13, Equations 1.7 and 1.8], a $_{14}\phi_{13}$ -identity [13, Equation 2.29] and a similar higher parameter identity [13, Equation 2.30], all due to Verma and Jain, and several similar identities found by McLaughlin and Zimmer [11]. However appropriate choices of substitutions are necessary. For instance, making the substitutions $a = q$, $x = z$, $y = z^{-1}$ in [13, Equation 1.7], letting $N \rightarrow \infty$ and differentiating twice with respect to z as in the earlier proofs leads us to the series

$$\sum_{n=0}^{\infty} \frac{(q; q^3)_n (q; q)_{n-1}}{(1 - q^n)(q; q)_{2n}} q^n$$

on the right hand side which is new. But, we have observed that making the substitutions $a \rightarrow 1$, $c = q$, $d = q^2$, $x = z$, $y = z^{-1}$ in [13, Equation 2.29] leads us again to the identity in Theorem 1.1. Further, multi-series generalizations of the $_{10}\phi_9$ -identities and the $_{14}\phi_{13}$ -identities due to Verma and Jain [13, Equations 4.1, 4.3, 4.4, 4.5] could lead to generalized spt-function identities.

3. There is resemblance and symmetry between our newfound spt-identities in Theorems 1.1 and 1.2 and those by the previous authors. For instance, upon bringing the lone product $(q; q)_{\infty}$ in our identity in Theorem 1.2 to the left hand side, the first sum on the right hand side of our identity and that of Andrews, Dixit

and Yee in Equation (1.3) are the same, more specifically the generating function for $np(n)$. And, the second sum in our identity can be obtained by substituting $q \rightarrow q^{\frac{3}{2}}$ in the second sum in their identity. It calls for the examination of the underlying theory behind this resemblance more closely.

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